Project Report: Splunk SIEM Integration with Sophos Central

1. Introduction

In today's cybersecurity landscape, organizations rely on Security Information and Event Management (SIEM) solutions to provide centralized visibility, detect threats, and automate alerts. This project demonstrates the deployment of Splunk Enterprise SIEM and its integration with Sophos Central. The goal was to collect and analyze security events, visualize trends using dashboards, and configure alerting for high-severity events.

2. Setup and Configuration

2.1 Splunk Installation

- Platform: (Windows/Linux)
- Installed Splunk Enterprise Trial from splunk.com.
- Accessed Splunk Web via https://localhost:8000.
- Verified successful login with the admin account.

2.2 Sophos Central API Credentials

- Registered for a Sophos Central Trial account.
- Created new API credentials in Global Settings → API Credentials Management.
- Captured the following:
  - Client ID
  - Client Secret
  - Tenant ID
  - API Region URL

2.3 SIEM Integration Script

- Downloaded the Sophos SIEM Integration Script from GitHub.
- Configured config.ini with API credentials and syslog settings.
- Set output format to CEF and pointed logs to Splunk syslog port (514).

2.4 Splunk Syslog Configuration

- Configured UDP 514 input in Splunk Web under Settings → Data Inputs.
- Assigned syslog sourcetype for incoming events.
- Confirmed logs were being ingested successfully.

3. Results

3.1 Dashboards and Reports

Created two dashboards in Splunk using SPL (Search Processing Language):

- 1. Blocked Malware Events

```
index=* sourcetype="syslog" "Event::Threat"
| stats count by deviceUserName, filePath, severity
| sort - count
```

- 2. Top Users with Security Alerts

```
index=* sourcetype="syslog" "Event::Alert"
 | stats count by deviceUserName
 | sort - count
```

3.2 Alerts

- Configured Splunk alert:
  Trigger: When severity = high in Sophos Threat events.
  Action: Send email notification to security admin.

4. Challenges

- API authentication: Required careful setup of Client ID/Secret.
- Syslog ingestion: Needed to allow Splunk to bind to port 514.
- Event parsing: Ensuring logs were correctly recognized in CEF format.

5. Conclusion

This project successfully demonstrated how to deploy Splunk Enterprise and integrate it with Sophos Central for centralized log collection and analysis. The integration provided:

- Real-time visibility into malware and alert events.
- Dashboards for quick security insights.
- Automated alerting for high-severity threats.

Such a setup strengthens an organization's security posture by enabling proactive monitoring and rapid response.

✅ Deliverables included in this report:

- Splunk installation and setup.
- Sophos API configuration.
- Logs ingested in Splunk.
- Dashboards and reports created.
- Alert rule tested successfully.